Wieferich Primes and the Exact Growth of Multiplicative Orders Modulo Prime Powers

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Abstract

OEIS A001220 lists odd primes p for which $p^2 \mid 2^{p-1}-1$. We prove a complete formula for the multiplicative order of 2 modulo p^k in terms of one p-adic valuation. It characterizes the Wieferich condition by a plateau in the order sequence and, more generally, characterizes exactly which powers p^k are Fermat pseudoprimes to base 2. This is a self-contained exposition of a known order-lifting principle, not a new result on the number of Wieferich primes.

The OEIS question and notation

The entry A001220 consists of primes p with $2^{p-1} \equiv 1 \pmod{p^2}$ (known examples include 1093 and 3511). Whether there are infinitely many is unresolved. For each odd prime p, set

$$d = \operatorname{ord}_p(2), \qquad t = v_p(2^d - 1),$$

where $d \mid p-1$ is the multiplicative order and v_p is the exponent of p in a nonzero integer. Necessarily $t \geq 1$.

Lemma 1 (Elementary exponent lifting)**.** *If p is odd, $p \mid x-1$, $x > 1$, and $h \geq 1$, then*

$$\boxed{v_p(x^h - 1) = v_p(x-1) + v_p(h).}$$

Proof. Write $x = 1 + p^s u$, with $s \geq 1$ and $p \nmid u$. If $p \nmid h$, the binomial expansion of $x^h - 1$ starts with $hp^s u$; every subsequent term is divisible by p^{s+1}, so its valuation is exactly s. For the exponent p, the first nonconstant term of $x^p - 1$ is $p^{s+1}u$. All intermediate binomial coefficients are divisible by p, and their terms have valuation at least $1 + 2s \geq s + 2$; the last term has valuation $ps \geq s + 2$, since $p \geq 3$. Thus $v_p(x^p - 1) = s + 1$. Repeatedly raising to the p-th power and then to the p-coprime part of h gives the assertion. □

Theorem 1 (The order staircase and pseudoprime powers)**.** *Let p be an odd prime, with d, t as above. For every integer $k \geq 1$,*

$$\boxed{\operatorname{ord}_{p^k}(2) = d\, p^{\max(0,k-t)}.}$$

Furthermore, the following are equivalent:

1. *p is a Wieferich prime (that is, $p \in$ A001220);*
2. *$t \geq 2$;*
3. *$\operatorname{ord}_{p^2}(2) = \operatorname{ord}_p(2)$;*
4. *p^2 is a Fermat pseudoprime to base 2: $2^{p^2-1} \equiv 1 \pmod{p^2}$.*

More generally, for each $k \geq 2$, the composite number p^k is a base-2 Fermat pseudoprime if and only if *$k \leq t$.*

Proof. If $2^n \equiv 1 \pmod{p^k}$, then in particular $d \mid n$, so write $n = dh$. Lemma 1 with $x = 2^d$ gives

$$v_p(2^n - 1) = t + v_p(h).$$

The least such positive n with valuation at least k occurs at $h = p^{\max(0,k-t)}$. This proves the order formula.

Since $d \mid p-1$ and $p \nmid (p-1)/d$, the same lemma yields

$$v_p(2^{p-1} - 1) = v_p\left((2^d)^{(p-1)/d} - 1\right) = t.$$

Consequently the defining Wieferich congruence is exactly $t \geq 2$, which by the order formula is equivalent to the order plateau from p to p^2.

Finally $d \mid p-1 \mid p^k - 1$ and $p \nmid (p^k-1)/d$; another application of the lemma gives

$$v_p(2^{p^k-1} - 1) = t.$$

Thus $p^k \mid 2^{p^k-1} - 1$ holds precisely when $k \leq t$. For $k = 2$ this yields the remaining equivalence. □

Examples and interpretation

The order is constant through modulus p^t and then increases by a factor p at each subsequent level. Small checks illustrate the distinction:

p	$d = \mathrm{ord}_p(2)$	t	$\mathrm{ord}_{p^3}(2)$
3	2	1	18
5	4	1	100
1093	364	2	$364 \cdot 1093$
3511	1755	2	$1755 \cdot 3511$

In particular 1093^2 and 3511^2 are base-2 Fermat pseudoprimes, whereas their cubes are not. These assertions follow from exact valuations, not from the empirical assertion that only two Wieferich primes are currently known.

Scope and provenance. The order-lifting mechanism is classical; the equivalence for p^2 appears in OEIS and in a public MathOverflow explanation. This note provides a derivation for all prime powers. It neither finds new Wieferich primes nor resolves their infinitude, and has not undergone independent human peer review.

References

[1] OEIS Foundation, A001220, Wieferich primes; see also A282552 (differences between orders modulo p and p^2), accessed November 2026.

[2] O. Gorodetsky, answer to "Let b be an odd integer greater than 1 ...", *MathOverflow*, 16 August 2022; mathoverflow.net/questions/428624. (Includes the p^2 equivalence; a comment there describes the general staircase.)